

Finite radius mechanism of the eta 0.35 axial event residual

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1 Result and scope

The nonmonotonic scaled advance at $\eta = 0.35$, $m = 3$ is reproduced by an independent finite angular Airy integral and explained by the higher order terms of the same smooth peak and half-height event. The fourth order event series does not include enough terms to locate this minimum. Its non-fitted extension gives $c_5 = 3.36680300$ and $c_6 = 13.35298604$, followed by smaller corrections at the radii in question. The fourteenth order prediction locates the minimum at $R_0 = 36.37128$; the full Fresnel integral gives 36.37127, and the manuscript's reduced R-S integral gives 36.37130. The last two digits are numerical estimates, not certified interval endpoints.

The integral identities and source-extension bound are analytic results. The reported event locations, absence of a branch change over the tested range and finite-order error envelopes are numerical evidence. The sampled-radius checks do not establish an interval proof of global minimum uniqueness or a result for other apertures, orders or damping parameters.

2 Model and event definition

The study retains the original radial envelope, the fixed- η path $\alpha = \eta/(2\sqrt{R_0})$, $m = 1$ as the reference, $P = 1$, the proxy aperture $b_m = j'_{m,1}$, and the first stationary maximum in $-3 \leq \tau \leq 5$. The crossing is the nearest preceding rising half-height. No physical zero-contour root is selected in this observable. The quantity being explained is $R_0\Delta$, not the error of a fitted coefficient.

$$\zeta = 2\sqrt{R_0} + \tau/\sqrt{R_0}, \quad R = \rho/w = 2x/\zeta.$$

Apart from common phases and physical constants, the Fresnel radial integral is

$$I_q(R, \zeta) = \int_0^\infty s \operatorname{Ai}(R_0 - s) e^{\alpha(R_0 - s)} e^{is^2/(2\zeta)} J_q(Rs/\zeta) ds.$$

The reduced R-S model replaces ζ in the phase and Bessel argument by $d = \sqrt{\zeta^2 + (R/kw)^2}$ and multiplies the integral by ζ/d^2 . The values used are $w = 0.08$ mm, $\lambda = 0.0006328$ mm and $kw = 2\pi w/\lambda$.

3 Exact compact reduction

3.1 The Airy convolution

For $\alpha > 0$ define the one-dimensional Fresnel propagation of the apodized Airy function:

$$F(y, \zeta) = \frac{1}{\sqrt{2\pi i \zeta}} \int_{-\infty}^\infty \operatorname{Ai}(t) e^{\alpha t} e^{i(y-t)^2/(2\zeta)} dt.$$

The initial profile has Fourier transform $\exp[(\alpha - ik)^3/3]$. Multiplication by $\exp(-i\zeta k^2/2)$ and completion of the cubic phase give

$$F(y, \zeta) = \text{Ai}(y - \zeta^2/4 + i\alpha\zeta)e^{\Phi(y, \zeta)},$$

$$\Phi(y, \zeta) = \alpha(y - \zeta^2/2) + i[\zeta(y + \alpha^2)/2 - \zeta^3/12].$$

This is the finite-energy Airy solution of Siviloglou and Christodoulides, Eq. (5) [1]. Setting $t = R_0 - s$, $a = R \cos \theta$ and $y = R_0 - a$ converts the full-line source integral without its factor s to $\sqrt{2\pi i\zeta} e^{-ia^2/(2\zeta)} F(R_0 - a, \zeta)$. Applying $i\zeta \partial_a$ restores the factor s . The sign follows directly by differentiating $\exp(-ias/\zeta)$.

For integer q , the Bessel angular identity [2] is

$$J_q(v) = \frac{i^q}{2\pi} \int_0^{2\pi} e^{-iv \cos \theta} e^{iq\theta} d\theta.$$

Let $X = R_0 - \zeta^2/4 + i\alpha\zeta$. After removing a nonzero common factor and normalizing by $\zeta^2/2$, the angular field is

$$\mathcal{H}_q = \frac{1}{2\pi} \int_0^{2\pi} e^{iq\theta} G(a, \zeta) \left[B(a, \zeta) \text{Ai}(X - a) - \frac{2i}{\zeta} \text{Ai}'(X - a) \right] d\theta,$$

$$G(a, \zeta) = e^{-i\zeta a/2 - \alpha a - ia^2/(2\zeta)}, \quad B(a, \zeta) = 1 - \frac{2i\alpha}{\zeta} + \frac{2a}{\zeta^2}.$$

The normalized proxy signal is therefore

$$\tilde{Q}_m = \frac{V e^{-2\eta\tau\epsilon - \eta\tau^2\epsilon^3/2}}{I_{0,m}} \int_0^{b_m} x (|\mathcal{H}_{m-1}|^2 - |\mathcal{H}_{m+1}|^2) dx,$$

where $\epsilon = R_0^{-1/2}$, $V = 1 + \tau\epsilon^2/2$ and $I_{0,m} = 2mJ_m^2(b_m)$. This normalization differs from Q_m by a positive factor independent of τ , so it leaves peaks and half-height events unchanged. The V and exponential factors must remain during event extraction.

For reduced R-S propagation, use d instead of ζ inside $\mathcal{H}_q$. Since d depends on x , its signal prefactor must be kept inside the x integral:

$$W_{\text{RS}}(x) = \frac{d\epsilon}{2} e^{-2\eta\tau\epsilon - \eta\tau^2\epsilon^3/2 - \alpha(R/kw)^2},$$

$$\tilde{Q}_m^{\text{RS}} = \frac{1}{I_{0,m}} \int_0^{b_m} x W_{\text{RS}}(x) (|\mathcal{H}_{m-1}(d)|^2 - |\mathcal{H}_{m+1}(d)|^2) dx.$$

This expression retains the exact geometry of the manuscript's reduced kernel. It is not a full Maxwell or unexpanded Rayleigh-Sommerfeld calculation. The angular identity sums the complete field coherently and introduces no arbitrary spectral partition.

3.2 Controlled negative-source extension

Replacing the physical interval $s \geq 0$ by the full real line adds an integral over $s = -v < 0$. Since $|J_q(t)| \leq 1$ for real t and integer q , the positive-axis Airy bound [3] and convexity of $(R_0 + v)^{3/2}$ give, for $\alpha < \sqrt{R_0}$,

$$|\delta I_q| \leq \frac{e^{\alpha R_0 - 2R_0^{3/2}/3}}{2\sqrt{\pi} R_0^{1/4} (\sqrt{R_0} - \alpha)^2}.$$

Indeed, $\text{Ai}(u) \leq e^{-2u^{3/2}/3}/(2\sqrt{\pi}u^{1/4})$ for $u > 0$, and $2(R_0 + v)^{3/2}/3 \geq 2R_0^{3/2}/3 + \sqrt{R_0}v$. The remaining integral is $\int_0^\infty ve^{-(\sqrt{R_0}-\alpha)v}dv$. The resulting dimensionless source-integral bounds are 1.1541×10^{-36} at $R_0 = 24$, 4.8580×10^{-76} at 40 and 2.433×10^{-275} at 96. These are bounds before the known propagation prefactor, not estimates obtained from cancellation. This artificial extension cannot account for the observed 10^{-2} changes in $R_0\Delta$.

4 Coefficients and event recursion

The angular formula admits direct expansion around $\epsilon = 0$. In the Fresnel case,

$$a = \frac{\epsilon x \cos \theta}{V}, \quad X = -\tau + i\eta + \epsilon^2(-\tau^2/4 + i\eta\tau/2), \quad \zeta a/2 = x \cos \theta.$$

Expand V^{-1} geometrically, the two remaining exponentials by power-series multiplication, and $\text{Ai}(X - a)$ about $-\tau + i\eta$. Its derivatives satisfy $A^{(k)} = X A^{(k-2)} + (k-2)A^{(k-3)}$ for $k \geq 3$, with $A'' = XA$. The field coefficients $H_{q,n}$ then determine every intensity coefficient by the convolution $\sum_{j=0}^n H_{q,j}H_{q,n-j}^*$. The common signal prefactor is convolved at the same total order. There is no truncation by derivative number followed by uncontrolled squaring.

Write $Q_m(\tau, \epsilon) = \sum_n q_{m,n}(\tau)\epsilon^n$, $p_m = p_0 + \sum_{n \geq 1} p_{m,n}\epsilon^n$ and $h_m = h_0 + \sum_{n \geq 1} h_{m,n}\epsilon^n$. For a simple peak and rising crossing, coefficient extraction gives

$$p_{m,n} = -\frac{[\epsilon^n] \partial_\tau Q_m(p_0 + \sum_{j < n} p_{m,j}\epsilon^j, \epsilon)}{q_{m,0}''(p_0)},$$

$$h_{m,n} = \frac{\frac{1}{2}[\epsilon^n] Q_m(p_0 + \sum_{j < n} p_{m,j}\epsilon^j, \epsilon) - [\epsilon^n] Q_m(h_0 + \sum_{j < n} h_{m,j}\epsilon^j, \epsilon)}{q_{m,0}'(h_0)}.$$

The unknown $p_{m,n}$ does not enter the peak height at this order because $q_{m,0}'(p_0) = 0$. This proves the recursion under its nondegeneracy assumptions. The coefficients $c_n = h_{1,n} - h_{3,n}$ are fixed by the model and event definition. They do not depend on a chosen set of propagation radii. The normalized full-line signal is analytic locally in ϵ where $V \neq 0$; the implicit function theorem supplies local analytic events while their denominators remain nonzero. The source extension contributes a beyond-all-orders term to the physical half-line integral. These local facts alone do not certify a global convergence radius for the event series.

n	c_n	n	c_n
2	4.41467828893	3	-1.44016807989
4	2.58765834920	5	3.36680299947
6	13.3529860411	7	21.1128977205
8	-49.8888463954	9	91.5175673968
10	237.809936151	11	-385.651361911
12	738.534340134	13	-809.692872335
14	3130.68031359	1	0 analytically

The computed c_1 is 1.2×10^{-15} . The first three nonzero coefficients reproduce the independent earlier fold derivation. Changing the Cauchy derivative-contour radius from 0.7 to 0.9 changes no coefficient through order 14 by more than 8.3×10^{-10} . Separate order-10 checks refine the radial and angular quadratures from 32/64 to 48/96 nodes. Cauchy extraction evaluates derivatives of an analytic formula; it does not fit propagated event values.

5 Why the minimum was missed

The scaled event series is $R_0\Delta = \sum_{n \geq 2} c_n R_0^{1-n/2}$. At fourth order it contains only $A_3 + B_3/\sqrt{R_0} + C_3/R_0$. Its stationary point is $(2C_3/|B_3|)^2 = 12.91361$, outside the observed intermediate-radius minimum. Adding the positive fifth- and sixth-order terms shifts the minimum toward the full calculation.

Event truncation	Predicted minimum R_0	Minimum $R_0\Delta$
4	12.91361	4.21429595
5	24.97050	4.25708537
6	34.41707	4.27232531
8	35.78792	4.27406412
10	36.41148	4.27452784
12	36.37256	4.27450296
14	36.37128	4.27450221
Full Fresnel	36.37127	4.27450221
Reduced R-S	36.37130	4.27450539

These are order controls without coefficient fitting. The high orders do not merely improve a regression statistic: they predict the location and depth of the minimum and the sign of its curvature. The decomposition below additionally identifies which operations generate the missing terms.

For each order n , split the new signal coefficient into (i) $H_0 H_n^* + H_n H_0^*$, (ii) lower-field products and the common prefactor, and (iii) inherited nonlinear terms from the two implicit event equations. Applying the linear crossing functional $[q_n(p_0)/2 - q_n(h_0)]/q'_0(h_0)$ to (i) and (ii), and assigning the remainder of h_n to (iii), gives the following exact algebraic grouping in this declared expansion convention.

Order	New field term	Lower products and prefactor	Nonlinear event terms	Total c_n
5	1.04031657	23.80479484	-21.47830841	3.36680300
6	22.98782712	54.38661509	-64.02145617	13.35298604

The large signed cancellations rule out interpreting c_5 or c_6 as the power carried by a separate branch. They are coherent finite-radius field corrections processed through peak normalization and a moving half-height. As an event control, forcing the target's normalization peak to the reference peak changes $R_0\Delta$ to 4.52744, 4.42926 and 4.36945 at 24, 40 and 96; the dip over these three points disappears. This artificial control isolates the importance of the peak condition but is not proposed as a replacement observable. Removing c_5 or c_6 from the fixed analytic series likewise displaces the minimum; all values are retained in `coefficient_decomposition.json`.

6 Predictions and independent numerical evidence

All coefficients were generated solely from the analytic identity; the generators never load propagation data. Predictions at the new radii were evaluated with these fixed coefficients. New controls are $R_0 = 20, 28, 36, 44, 48, 64, 80$; the report also recomputes the original radii and extends to 256. The maximum fourteenth-order error in $R_0\Delta$ against the full Fresnel identity on the new controls is 2.84×10^{-7} , attained at 20. Over sampled $24 \leq R_0 \leq 96$ it is below 7.1×10^{-8} . The small reduced R-S/Fresnel difference is retained separately, rather than absorbed into a fitted high-order term.

R_0	Reduced R-S $R_0\Delta$	Fourteenth-order Fresnel error in $R_0\Delta$	R-S minus Fresnel
24	4.286137374	$+7.02 \times 10^{-8}$	3.225×10^{-6}
36	4.274510642	-2.94×10^{-9}	3.178×10^{-6}
40	4.274925216	-2.94×10^{-9}	3.172×10^{-6}
48	4.277668295	-1.75×10^{-9}	3.166×10^{-6}
64	4.285437979	-4.95×10^{-10}	3.161×10^{-6}
96	4.299868386	-5.99×10^{-11}	3.161×10^{-6}

The angular implementation uses Gauss-Legendre integration in x and periodic trapezoidal integration in angle. The independent positive-source implementation evaluates the original oscillatory radial integral by composite Gauss-Legendre quadrature, with exact reduced R-S geometry and without a spectral transform, phase expansion, radial interpolation or axial spline. Separate tests halve the source panels and extend the exponential cutoff. Refining panels from 0.5 to 0.25 and cutoff from 22 to 26 changes the scaled advance by at most 4.64×10^{-10} . The finest direct-source results differ from the compact reduced R-S evaluation by at most 1.41×10^{-12} in $R_0\Delta$. The arbitrary precision implementation uses a third angular evaluation: an Airy Taylor series in a and analytic angular moments expressed as Bessel derivatives. It uses mpmath at 35 and 50 decimal digits, 20/28 radial nodes and 36/48 angular-series terms. At the two event locations for each of $m = 1, 3$ and $R_0 = 24, 40, 96$, the maximum signal difference between these precision settings is 6.0×10^{-33} ; the double-precision angular implementation differs by at most 1.45×10^{-15} . These are measured numerical differences, not interval-certified bounds.

The first peaks and preceding rising crossings continue smoothly across the 16 tested radii from 16 to 256. The minimum crossing slope of the normalized signal is 0.1035; the minimum magnitude of the selected peak curvature is 0.2561. All detected later maxima are retained in `event_continuation.csv`. The selected first maxima remain separated from them. Thus the tested minimum is not associated with a detected event switch, peak degeneracy or a selected-zero jump.

7 Error accounting and interpretation

The signed residual admits the exact bookkeeping identity

$$\Delta_{\text{RS}} - \Delta_4 = (\Delta_{\text{RS}} - \Delta_{\text{F}}) + \sum_{n=5}^{14} c_n R_0^{-n/2} + (\Delta_{\text{F}} - \Delta_{14}).$$

The first term is the model geometry correction, the second is the analytic high-order event contribution and the third is the retained series remainder. The source-extension bound and numerical evaluation error are reported separately. This identity closes the residual without allocating coherent powers to arbitrary spectral windows.

Contribution	Evidence and scale	Interpretation
Source extension	Analytic bound 1.16×10^{-36} on I_q at 24	Rigorous positive-source extension control
Angular and radial quadratures	48/96 to 72/128 nodes: scaled-advance change below 6.2×10^{-14}	Measured numerical convergence
Independent source quadrature	Source refinement change below 4.64×10^{-10} in scaled advance	Direct check of the original integral

Contribution	Evidence and scale	Interpretation
Arbitrary precision signal evaluation	Maximum double/50-digit difference 1.45×10^{-15}	Independent Airy/Bessel implementation
Fourteenth-order remainder	Below 7.1×10^{-8} in sampled $R_0\Delta$, 24–96	Finite-order approximation, not quadrature noise
Reduced R-S versus Fresnel	$3.16\text{--}3.23 \times 10^{-6}$ in $R_0\Delta$, 24–96	Separate kernel geometry correction
Observed dip and recovery	24 to 40: -0.01121216 ; 40 to 96: $+0.02494317$	Resolved finite-radius behavior

The mechanism supported here is a smooth, coherent high-order correction to the fixed-proxy signal and its peak-normalized event. It does not require a root change or a numerical artifact, and the independently derived terms quantitatively account for the minimum. A unique division of this correction into inward, outward and endpoint powers remains representation dependent; such a division is neither claimed nor needed for this residual closure.

8 Reproduction and provenance

Run the scripts from the `residual_model` directory with Python, NumPy, SciPy, mpmath and Matplotlib. The full recipe is in `RUN_REPRODUCTION.sh`. `compact_airy.py` and `compact_rs.py` implement the identities; `analytic_series.py` generates the coefficients; `study.py` and the N14 driver generate the radius predictions; `direct_check.py` and `direct_events.py` evaluate the original positive-source integral; `high_precision.py` uses arbitrary precision angular moments; `decomposition.py` gives the signed coefficient grouping. JSON and CSV files retain parameters, event roots, slopes, curvatures and control results. Source-code and output hashes are supplied with the verification package.

Code and output checksums are listed in `DISTRIBUTION_MANIFEST.json` at the package root.

9 References

1. G. A. Siviloglou and D. N. Christodoulides, Accelerating finite energy Airy beams, *Optics Letters* 32, 979–981 (2007), doi: 10.1364/OL.32.000979. The Fourier spectrum in Eq. (3) and propagated field in Eq. (5) supply the one-dimensional convolution identity.
2. NIST Digital Library of Mathematical Functions, 10.9.2 and 9.5, Bessel angular representation and Airy contour identities. Accessed 4 October 2026. These provide the exact transform identities.
3. NIST Digital Library of Mathematical Functions, 9.7.15, positive-real Airy inequality. Accessed 4 October 2026. Used for the explicit extension bound.
4. N. K. Efremidis and D. N. Christodoulides, Abruptly autofocusing waves, *Optics Letters* 35, 4045–4047 (2010), doi: 10.1364/OL.35.004045. Establishes circular Airy autofocusing.
5. T. Geng, M. Li and H. Guo, Orbit-induced localized spin angular momentum of vector circular Airy vortex beam in the paraxial regime, *Optics Express* 29, 14069–14077 (2021). Publisher record. Provides the radial-channel propagation reduction.
6. C. Chester, B. Friedman and F. Ursell, An extension of the method of steepest descents, *Proceedings of the Cambridge Philosophical Society* 53, 599–611 (1957), doi: 10.1017/S0305004100032655. Treats coalescing saddles and uniform Airy expansions. Finite truncation alone does not bound the present event functional.

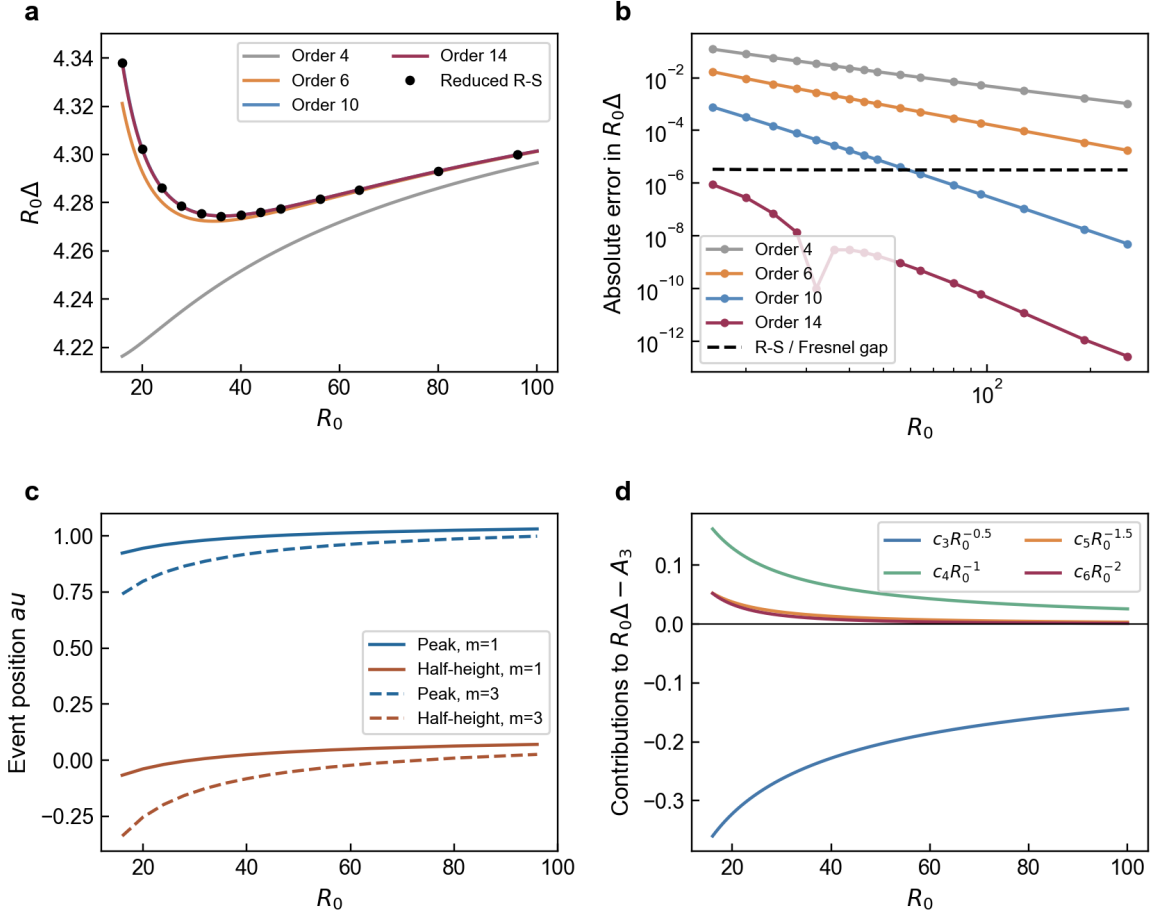


Figure 1: Mechanism evidence. (a) Non-fitted event expansions and full reduced R-S values. (b) Truncation errors against the full Fresnel identity and the separately retained geometry correction. (c) First peak and rising half-height continuation. (d) Signed lower-order contributions to the scaled advance.